

AI/BigData 인텐시브 과정

강사장철원

Section 0

코스소개

DAY1

DAY2

DAY3

DAY4

DAY5

데이터 분석을 위한 기초 수학

데이터 집계 및 시각화

DAY6

DAY7

DAY8

DAY9

DAY10

미니
프로젝트

확률분포, AB테스트

DAY11

DAY12

DAY13

DAY14

DAY15

머신러닝

DAY16

DAY17

DAY18

DAY19

DAY20

딥러닝

파이널 프로젝트

□ 확률변수

□ 조건부확률

□ 평균, 분산, 표준편차

□ 모집단과 표본

□ 공분산/상관계수

AI/BigData 인텐시브 과정

Section 1. 기초 통계

Section 1-1. 확률 변수

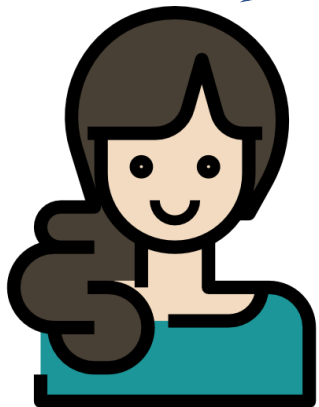
확률 변수(random variable)

확률 변수(random variable): 확률 현상을 통해 값이 확률적으로 정해지는 변수

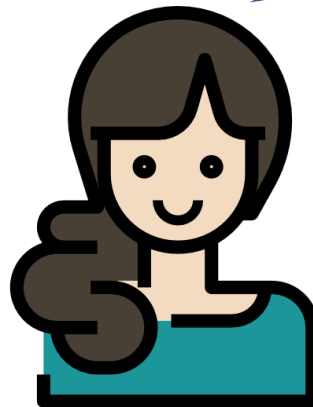
확률 현상

발생 가능한 결과들은 알지만 가능한 결과들 중 어떤 결과가 나올지는 모르는 현상

동전 던지면 앞면 아니면 뒷면이
나오는건 확실해



그런데 앞면/뒷면 중 뭐가 나올지는 몰라!



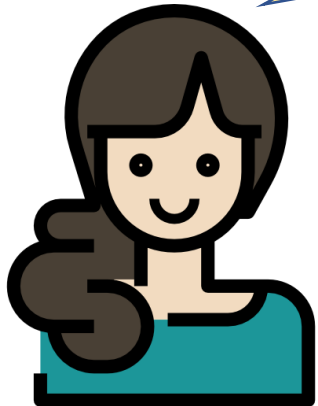
확률

이론적 확률 vs 경험적 확률

이론적 확률 vs 경험적 확률

이론적 확률 – 이론적(theoretical), 미래(future)

동전을 던지면 앞면이 나올 확률은 ?



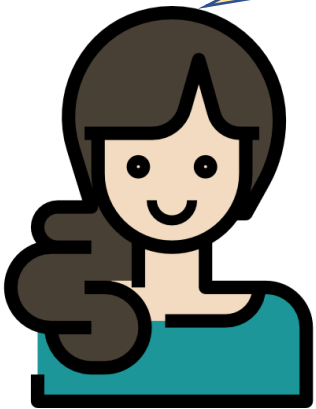
$$\text{확률} = \frac{\text{관심 대상}}{\text{나올수 있는 결과 갯수}}$$

$$\text{앞면이 나올 확률} = \frac{\text{앞면}}{\text{앞면} + \text{뒷면}} = \frac{1}{2} = 50\%$$

이론적 확률 vs 경험적 확률

경험적 확률 – 경험적(empirical), 과거(past)

동전을 100번 던졌더니
앞면이 60번, 뒷면이 40번 나왔어



$$\text{경험적 확률} = \frac{\text{관심 대상}}{\text{전체 데이터 수}}$$

$$\text{앞면의 비율} = \frac{60}{100} = \frac{3}{5} = 60\%$$

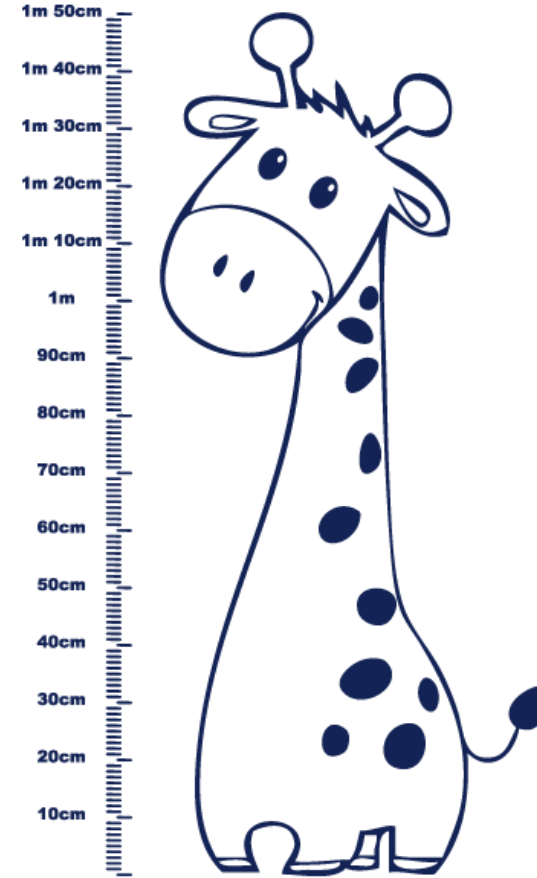
확률 변수 vs 상수

10

π
3,14
15926

7

x



확률의 정의

확률: 어떤 사건이 발생할 가능성을 수치화 시킨 것으로 0과 1사이의 값을 가짐

- 동전을 던졌을 때 앞면이 나올 확률은?
- 주사위를 던졌을 때 6이 나올 확률은?
- 게임 유저가 다음 주에 접속할 확률은?

확률의 공리적 정의

1) $0 \leq P(A) \leq 1$

2) $P(\Omega) = 1$

3) 만약 $A_1, A_2, \dots$ 가 상호 배반 사건이면, $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

표본 공간(sample space)

- 동전 던졌을 때, $\Omega = \{\text{앞면}, \text{뒷면}\}$
- 주사위를 던졌을 때, $\Omega = \{1, 2, 3, 4, 5, 6\}$
- 상품권 추첨할 때, $\Omega = \{\text{당첨}, \text{미당첨}\}$

AI/BigData 인텐시브 과정

Section 1. 기초 통계

Section 1-2. 조건부 확률

확률의 독립(independent)

좌변과 우변의 값이 같으면 독립

$$P(A \cap B) = P(A)P(B)$$



사건 A와 사건 B가 동시에 발생할 확률



사건 A가 발생할 확률 X 사건 B가 발생할 확률

확률의 배반(disjoint)

$$P(A \cap B) = 0$$

사건 A와 사건 B가 동시에 발생할 확률이 0

조건부 확률

사건 Y와 사건X가 동시에 발생할 확률

$$P(Y|X) = \frac{P(Y \cap X)}{P(X)}$$

사건 X가 발생할 확률

사건 X가 발생했을 때, 사건 Y가 발생할 확률

조건부 확률

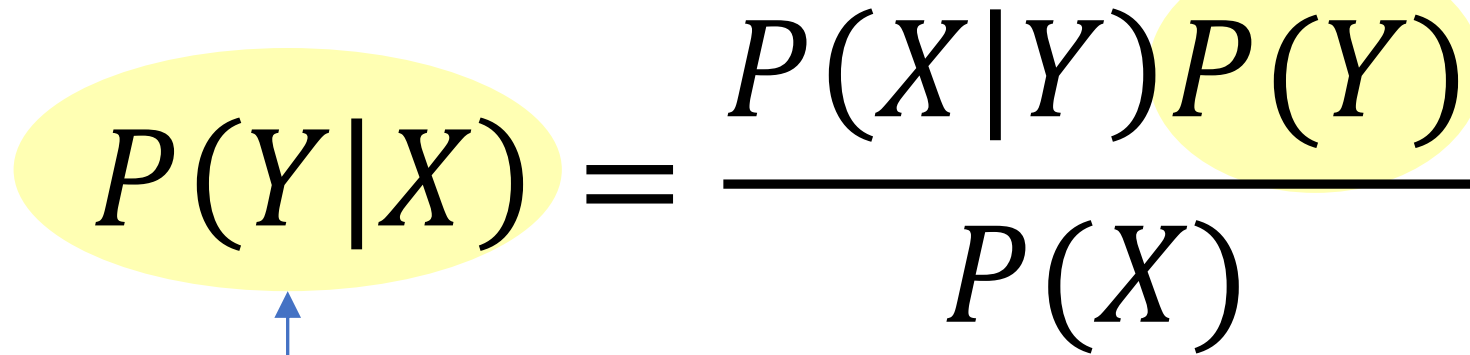
$$P(Y|X) = \frac{P(Y \cap X)}{P(X)}$$

$$P(Y \cap X) = P(Y|X)P(X) = P(X|Y)P(Y)$$

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

조건부 확률

사전확률(Prior Probability) : 이벤트 발생 전 확률



The diagram illustrates the components of Bayes' theorem. The equation $P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$ is shown. The term $P(Y|X)$ is enclosed in a yellow oval, with a blue arrow pointing up to it from the text '사후확률(Posterior Probability) : 이벤트 발생 후 확률'. The term $P(Y)$ in the numerator is also enclosed in a yellow oval, with a blue arrow pointing down to it from the text '사전확률(Prior Probability) : 이벤트 발생 전 확률'.

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

사후확률(Posterior Probability) : 이벤트 발생 후 확률

조건부 확률

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

X는 Y에 영향을 받지 않으므로 상수 취급

$$\propto P(X|Y)P(Y)$$

조건부 확률(X와 Y가 독립인 경우)

$$P(Y|X) = \frac{P(Y \cap X)}{P(X)} = \frac{P(Y)P(X)}{P(X)} = P(Y)$$

$$P(Y|X) = P(Y)$$

방의 개수(RM)가 7개 초과일 확률

$$P(\text{방의 개수} > 7) = ?$$

방의 개수(RM)가 7개 초과일 확률

```
In [1]: 1 import numpy as np
        2 import pandas as pd
```

```
In [2]: 1 df = pd.read_csv("../data/house_prices.csv", encoding='cp949')
```

```
In [3]: 1 df
```

Out[3]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36.2
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	9.67	22.4
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9.08	20.6
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5.64	23.9
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6.48	22.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	7.88	11.9

506 rows × 14 columns

방의 개수(RM)가 7개 초과일 확률

```
In [10]: ▶ 1 rm = df[df['RM'] > 7]
          2 rm
```

Out[10]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
2	0.02729	0.0	7.07	0	0.4690	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
4	0.06905	0.0	2.18	0	0.4580	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36.2
40	0.03359	75.0	2.95	0	0.4280	7.024	15.8	5.4011	3	252	18.3	395.62	1.98	34.9
55	0.01311	90.0	1.22	0	0.4030	7.249	21.9	8.6966	5	226	17.9	395.93	4.81	35.4
64	0.01951	17.5	1.38	0	0.4161	7.104	59.5	9.2229	3	216	18.6	393.24	8.05	33.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
364	3.47428	0.0	18.10	1	0.7180	8.780	82.9	1.9047	24	666	20.2	354.55	5.29	21.9
370	6.53876	0.0	18.10	1	0.6310	7.016	97.5	1.2024	24	666	20.2	392.05	2.96	50.0
375	19.60910	0.0	18.10	0	0.6710	7.313	97.9	1.3163	24	666	20.2	396.90	13.44	15.0
453	8.24809	0.0	18.10	0	0.7130	7.393	99.3	2.4527	24	666	20.2	375.87	16.74	17.8
482	5.73116	0.0	18.10	0	0.5320	7.061	77.0	3.4106	24	666	20.2	395.28	7.01	25.0

64 rows × 14 columns

```
In [11]: ▶ 1 n_rm = len(rm)
          2 print(n_rm)
```

64

```
In [13]: ▶ 1 n = len(df)
          2 print(n)
```

506

```
In [14]: ▶ 1 p = n_rm/n
          2 print(p)
```

0.12648221343873517

조건부 확률

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(\text{방 개수} > 7 \mid \text{학생 교사 비율} < 15) = \frac{P((\text{방 개수} > 7) \& (\text{학생 교사 비율} < 15))}{P(\text{학생 교사 비율} < 15)}$$

조건부 확률

$P(\text{학생 교사 비율} < 15) =$

```
In [22]: 1 ptratio = df[df['PTRATIO'] < 15]
        2 ptratio.head(10)
```

Out[22]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
142	3.32105	0.0	19.58	1	0.871	5.403	100.0	1.3216	5	403	14.7	396.90	26.82	13.4
143	4.09740	0.0	19.58	0	0.871	5.468	100.0	1.4118	5	403	14.7	396.90	26.42	15.6
144	2.77974	0.0	19.58	0	0.871	4.903	97.8	1.3459	5	403	14.7	396.90	29.29	11.8
145	2.37934	0.0	19.58	0	0.871	6.130	100.0	1.4191	5	403	14.7	172.91	27.80	13.8
146	2.15505	0.0	19.58	0	0.871	5.628	100.0	1.5166	5	403	14.7	169.27	16.65	15.6
147	2.36862	0.0	19.58	0	0.871	4.926	95.7	1.4608	5	403	14.7	391.71	29.53	14.6
148	2.33099	0.0	19.58	0	0.871	5.186	93.8	1.5296	5	403	14.7	356.99	28.32	17.8
149	2.73397	0.0	19.58	0	0.871	5.597	94.9	1.5257	5	403	14.7	351.85	21.45	15.4
150	1.65660	0.0	19.58	0	0.871	6.122	97.3	1.6180	5	403	14.7	372.80	14.10	21.5
151	1.49632	0.0	19.58	0	0.871	5.404	100.0	1.5916	5	403	14.7	341.60	13.28	19.6

```
In [30]: 1 p_ptratio = len(ptratio)/len(df)
        2 print(p_ptratio)
```

0.11462450592885376

조건부 확률

$$P((\text{방 개수} > 7) \& (\text{학생 교사 비율} < 15))$$

```
In [28]: ▶ 1 ptratio_rm = df[ (df['RM'] > 7) & (df['PTRATIO'] < 15) ]  
          2 ptratio_rm.head(10)
```

Out[28]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT	MEDV
161	1.46336	0.0	19.58	0	0.6050	7.489	90.8	1.9709	5	403	14.7	374.43	1.73	50.0
162	1.83377	0.0	19.58	1	0.6050	7.802	98.2	2.0407	5	403	14.7	389.61	1.92	50.0
163	1.51902	0.0	19.58	1	0.6050	8.375	93.9	2.1620	5	403	14.7	388.45	3.32	50.0
166	2.01019	0.0	19.58	0	0.6050	7.929	96.2	2.0459	5	403	14.7	369.30	3.70	50.0
195	0.01381	80.0	0.46	0	0.4220	7.875	32.0	5.6484	4	255	14.4	394.23	2.97	50.0
196	0.04011	80.0	1.52	0	0.4040	7.287	34.1	7.3090	2	329	12.6	396.90	4.08	33.3
197	0.04666	80.0	1.52	0	0.4040	7.107	36.6	7.3090	2	329	12.6	354.31	8.61	30.3
198	0.03768	80.0	1.52	0	0.4040	7.274	38.3	7.3090	2	329	12.6	392.20	6.62	34.6
202	0.02177	82.5	2.03	0	0.4150	7.610	15.7	6.2700	2	348	14.7	395.38	3.11	42.3
203	0.03510	95.0	2.68	0	0.4161	7.853	33.2	5.1180	4	224	14.7	392.78	3.81	48.5

```
In [31]: ▶ 1 p_ptratio_rm = len(ptratio_rm)/len(df)  
          2 print(p_ptratio_rm)
```

0.04940711462450593

조건부 확률

$$P(\text{방 개수} > 7 \mid \text{학생 교사 비율} < 15) = \frac{P((\text{방 개수} > 7) \& (\text{학생 교사 비율} < 15))}{P(\text{학생 교사 비율} < 15)}$$

In [33]:



```
1 p = p_ptratio_rm / p_ptratio
2 print(p)
```

0.43103448275862066

AI/BigData 인텐시브 과정

Section 1. 기초 통계

Section 1-3. 평균, 분산, 표준 편차

평균

$$\bar{X} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

$$= \frac{1}{n} \sum_{i=1}^n x_i$$

평균

1, 3, 5, 9, 12

$$\bar{x} = \frac{1 + 3 + 5 + 9 + 12}{5} = \frac{30}{5} = 6$$

위치 모수(location parameter)

분산 & 표준 편차

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

scale parameter

분산 & 표준 편차

$$1, 3, 5, 9, 12$$

$$\bar{x} = 6$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \frac{1}{5-1} \{(1-6)^2 + (3-6)^2 + (5-6)^2 + (9-6)^2 + (12-6)^2\}$$

$$= \frac{1}{4} \{25 + 9 + 1 + 9 + 36\} = \frac{1}{4} \times 80$$

$$= 20$$

$$s^2 = 20$$

$$s = \sqrt{20}$$

자유도(degree of freedom)의 개념

1, 3, 4, 6, x

자유도(degree of freedom)의 개념

1, 3, 4, 6, x



6

$$\bar{x} = 4$$

자유도(degree of freedom)의 개념

$$x_1, x_2, x_3, x_4, x_5$$

자유도(degree of freedom)의 개념

$$x_1, x_2, x_3, x_4, x_5$$

$$\bar{x} = 3$$

평균

```
In [37]: ▶ 1 np.mean(df['RM'])
```

```
Out[37]: 6.284634387351779
```

```
In [38]: ▶ 1 df['RM'].mean()
```

```
Out[38]: 6.284634387351779
```

```
In [54]: ▶ 1 n = len(df)
2 sum_x = 0
3 for x in df['RM']:
4     sum_x += x
5 mean_rm = sum_x/n
6 print(mean_rm)
```

```
6.284634387351787
```

$$\bar{X} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

분산, 표준 편차

In [39]: `1 np.var(df['RM'])`

Out[39]: 0.49269521612976297

In [41]: `1 np.std(df['RM'])`

Out[41]: 0.7019225143345689

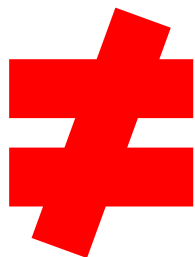
In [40]: `1 df['RM'].var()`

Out[40]: 0.49367085022110907

In [42]: `1 df['RM'].std()`

Out[42]: 0.7026171434153233

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$



$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

```
1 n = len(df)
2 ss = 0
3 for x in df['RM']:
4     ss += (x - mean_rm)**2
```

```
1 var_rm = ss/n
2 std_rm = var_rm**0.5
3 print(var_rm)
4 print(std_rm)
```

0.49269521612976347

0.7019225143345692

```
1 var_rm = ss/(n-1)
2 std_rm = var_rm**0.5
3 print(var_rm)
4 print(std_rm)
```

0.4936708502211095

0.7026171434153237

numpy, pandas 차이

```
In [65]: ▶ 1 df['RM'].var()
```

```
Out[65]: 0.49367085022110907
```

```
In [66]: ▶ 1 df['RM'].var(ddof=1)
```

```
Out[66]: 0.49367085022110907
```

```
In [67]: ▶ 1 df['RM'].var(ddof=0)
```

```
Out[67]: 0.49269521612976297
```

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

pandas.DataFrame.var

`DataFrame.var(axis=None, skipna=True, level=None, ddof=1, numeric_only=None, **kwargs)` [\[source\]](#)

Return unbiased variance over requested axis.

Normalized by N-1 by default. This can be changed using the ddof argument.

Parameters: **axis** : {index (0), columns (1)}

For Series this parameter is unused and defaults to 0.

skipna : bool, default True

Exclude NA/null values. If an entire row/column is NA, the result will be NA.

level : int or level name, default None

If the axis is a MultiIndex (hierarchical), count along a particular level, collapsing into a Series.

! Deprecated since version 1.3.0: The level keyword is deprecated. Use groupby instead.

ddof : int, default 1

Delta Degrees of Freedom. The divisor used in calculations is N - ddof, where N represents the number of elements.

numeric_only : bool, default None

Include only float, int, boolean columns. If None, will attempt to use everything, then use only numeric data. Not implemented for Series.

numpy, pandas 차이

```
In [68]: 1 df['RM'].std()
```

```
Out[68]: 0.7026171434153233
```

```
In [69]: 1 df['RM'].std(ddof=1)
```

```
Out[69]: 0.7026171434153233
```

```
In [70]: 1 df['RM'].std(ddof=0)
```

```
Out[70]: 0.7019225143345689
```

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

pandas.DataFrame.std

`DataFrame.std(axis=None, skipna=True, level=None, ddof=1, numeric_only=None, **kwargs)` [\[source\]](#)

Return sample standard deviation over requested axis.

Normalized by N-1 by default. This can be changed using the ddof argument.

Parameters: **axis** : {index (0), columns (1)}

For *Series* this parameter is unused and defaults to 0.

skipna : bool, default True

Exclude NA/null values. If an entire row/column is NA, the result will be NA.

level : int or level name, default None

If the axis is a MultiIndex (hierarchical), count along a particular level, collapsing into a *Series*.

! *Deprecated since version 1.3.0:* The level keyword is deprecated. Use *groupby* instead.

ddof : int, default 1

Delta Degrees of Freedom. The divisor used in calculations is N - ddof, where N represents the number of elements.

numeric_only : bool, default None

Include only float, int, boolean columns. If None, will attempt to use everything, then use only numeric data. Not implemented for *Series*.

numpy, pandas 차이

```
In [71]: 1 np.var(df['RM'])
```

```
Out[71]: 0.49269521612976297
```

```
In [72]: 1 np.var(df['RM'], ddof=0)
```

```
Out[72]: 0.49269521612976297
```

```
In [73]: 1 np.var(df['RM'], ddof=1)
```

```
Out[73]: 0.49367085022110907
```

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

numpy.var

`numpy.var(a, axis=None, dtype=None, out=None, ddof=0, keepdims=<no value>, *, where=<no value>)` [\[source\]](#)

Compute the variance along the specified axis.

Returns the variance of the array elements, a measure of the spread of a distribution. The variance is computed for the flattened array by default, otherwise over the specified axis.

Parameters: `a : array_like`

Array containing numbers whose variance is desired. If `a` is not an array, a conversion is attempted.

`axis : None or int or tuple of ints, optional`

Axis or axes along which the variance is computed. The default is to compute the variance of the flattened array.

! New in version 1.7.0.

If this is a tuple of ints, a variance is performed over multiple axes, instead of a single axis or all the axes as before.

`dtype : data-type, optional`

Type to use in computing the variance. For arrays of integer type the default is `float64`; for arrays of float types it is the same as the array type.

`out : ndarray, optional`

Alternate output array in which to place the result. It must have the same shape as the expected output, but the type is cast if necessary.

`ddof : int, optional`

"Delta Degrees of Freedom": the divisor used in the calculation is `N - ddof`, where `N` represents the number of elements. By default `ddof` is zero.

numpy, pandas 차이

```
In [74]: ▶ 1 np.std(df['RM'])
```

```
Out[74]: 0.7019225143345689
```

```
In [75]: ▶ 1 np.std(df['RM'], ddof=0)
```

```
Out[75]: 0.7019225143345689
```

```
In [76]: ▶ 1 np.std(df['RM'], ddof=1)
```

```
Out[76]: 0.7026171434153233
```

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

numpy.std

`numpy.std(a, axis=None, dtype=None, out=None, ddof=0, keepdims=<no value>, *, where=<no value>)` [\[source\]](#)

Compute the standard deviation along the specified axis.

Returns the standard deviation, a measure of the spread of a distribution, of the array elements. The standard deviation is computed for the flattened array by default, otherwise over the specified axis.

Parameters: `a` : *array_like*

Calculate the standard deviation of these values.

`axis` : *None or int or tuple of ints, optional*

Axis or axes along which the standard deviation is computed. The default is to compute the standard deviation of the flattened array.

! New in version 1.7.0.

If this is a tuple of ints, a standard deviation is performed over multiple axes, instead of a single axis or all the axes as before.

`dtype` : *dtype, optional*

Type to use in computing the standard deviation. For arrays of integer type the default is float64, for arrays of float types it is the same as the array type.

`out` : *ndarray, optional*

Alternative output array in which to place the result. It must have the same shape as the expected output but the type (of the calculated values) will be cast if necessary.

`ddof` : *int, optional*

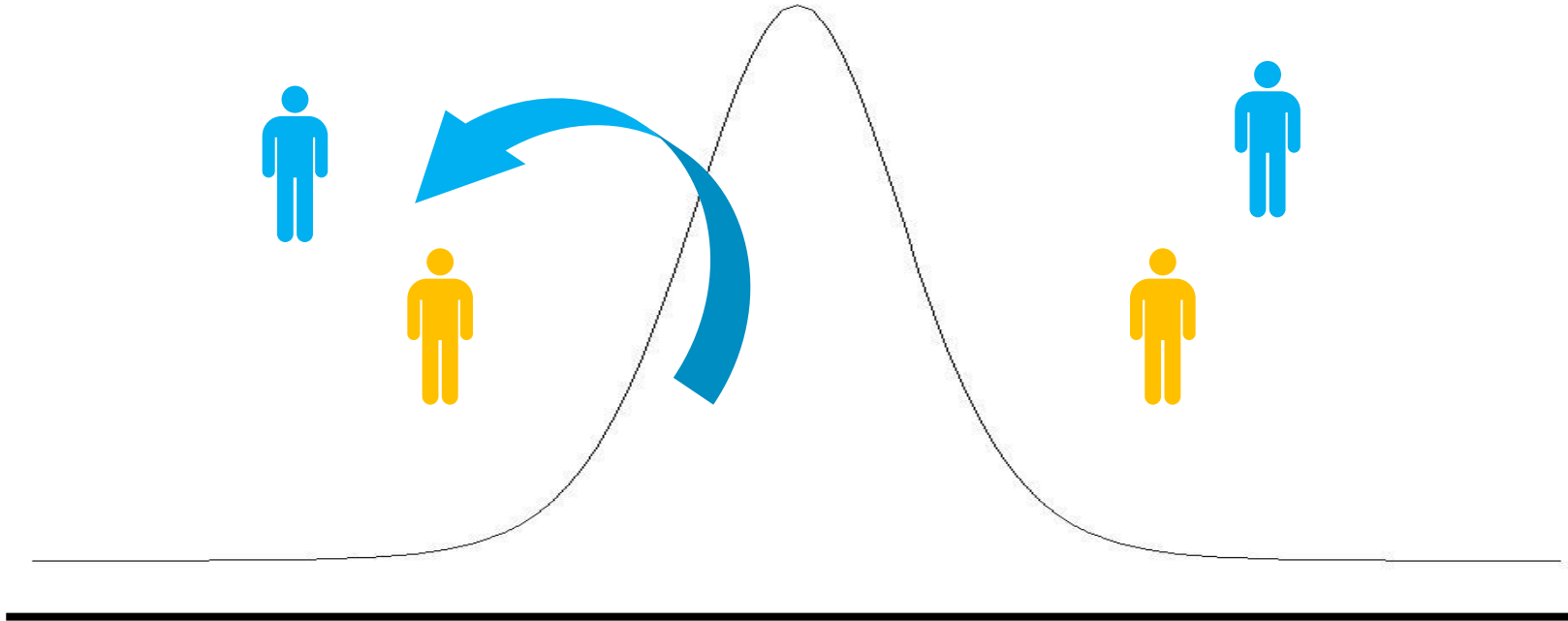
Means Delta Degrees of Freedom. The divisor used in calculations is `N - ddof`, where `N` represents the number of elements. **By default `ddof` is zero.**

AI/BigData 인텐시브 과정

Section 1. 기초 통계

Section 1-4. 모집단과 표본

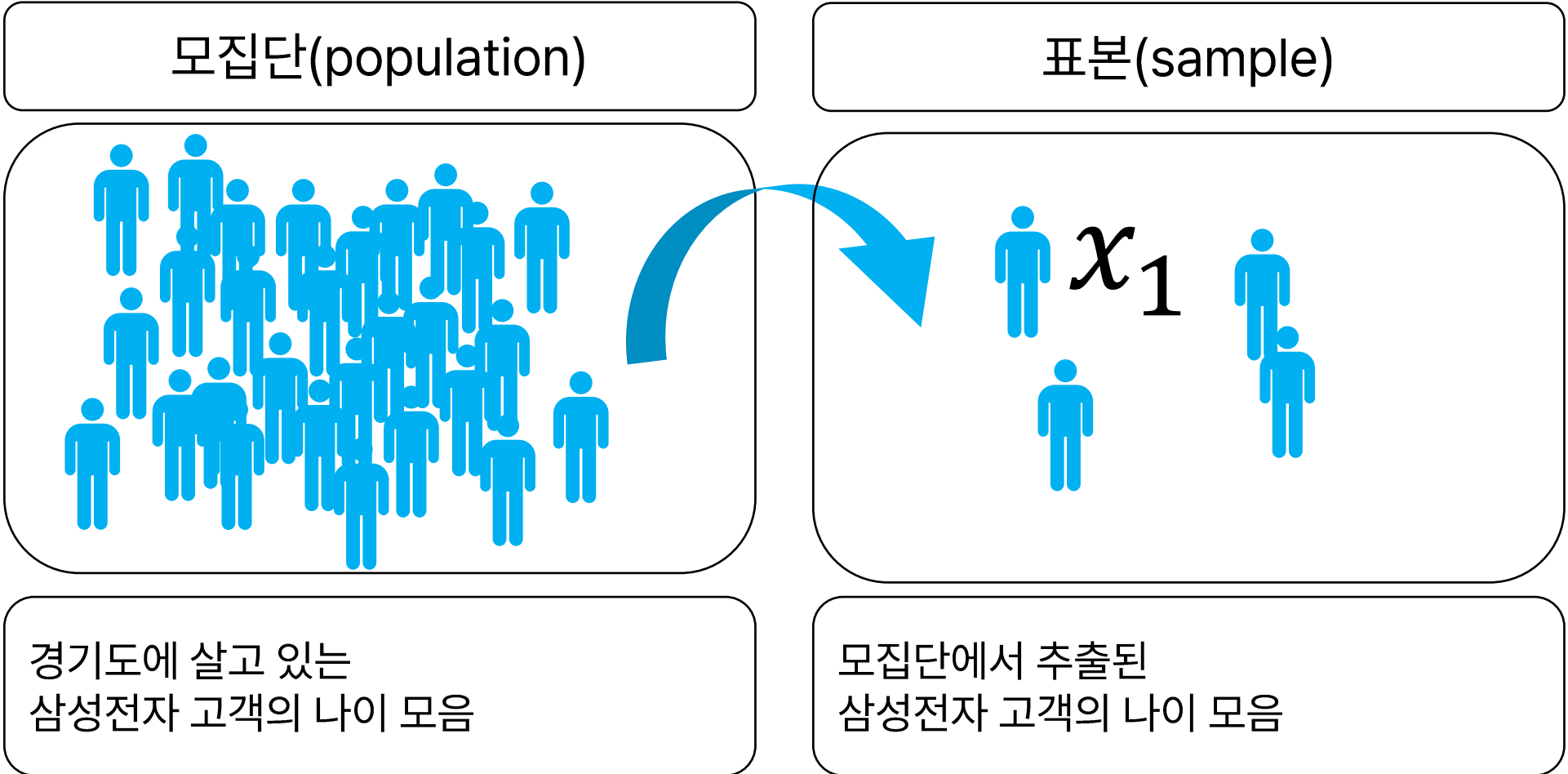
모집단(population)과 표본(sample)



모집단(population) : 관심대상 전체 집합

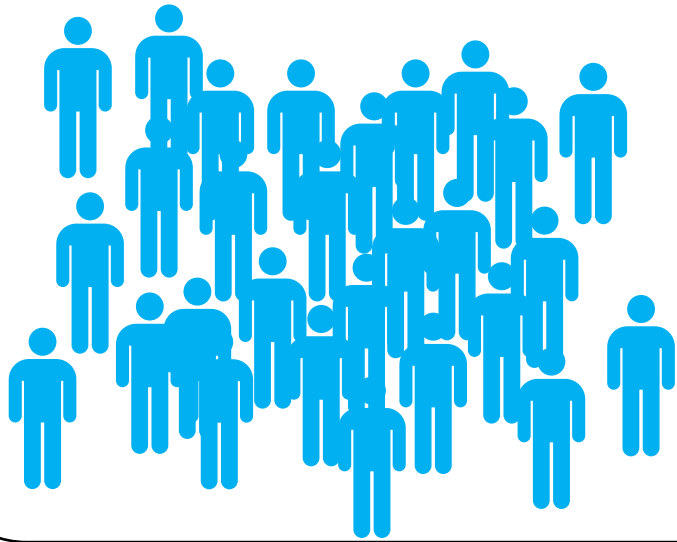
표본(sample) : 모집단의 부분 집합

모집단(population)과 표본(sample)



모집단(population)과 표본(sample)

모집단(population)



경기도에 살고 있는
삼성전자 고객의 나이 모음

모평균(population mean) = 모집단의 평균

$$\mu$$

모분산(population variance) = 모집단의 분산

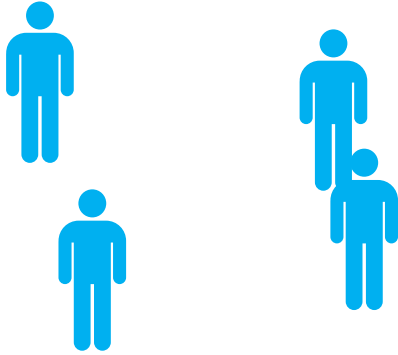
$$\sigma^2$$

모표준편차(population standard deviation) = 모집단의 표준편차

$$\sigma$$

모집단(population)과 표본(sample)

표본(sample)



모집단에서 추출된
삼성전자 고객의 나이 모음

표본 평균(sample mean) = 표본의 평균

$$\bar{x}$$

표본 분산(sample variance) = 표본의 분산

$$s^2$$

표본 표준편차(sample standard deviation) = 표본의 표준편차

$$s$$

모수(population parameter)

모수(population parameter)

= 모집단의 특성을 나타내는 대표값

ex) 모평균, 모분산, 모표준편차

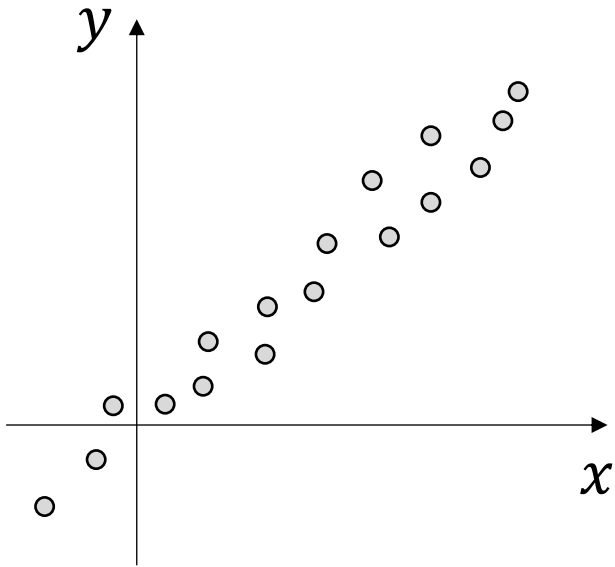
μ
 σ^2 σ

AI/BigData 인텐시브 과정

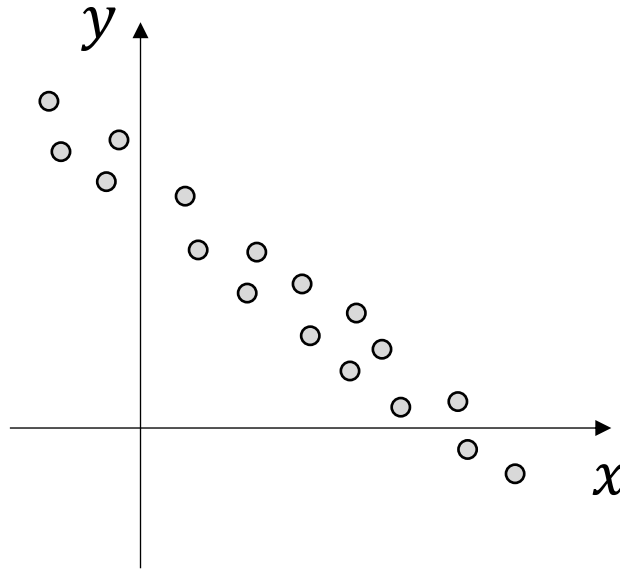
Section 1. 기초 통계

Section 1-5. 공분산과 상관계수

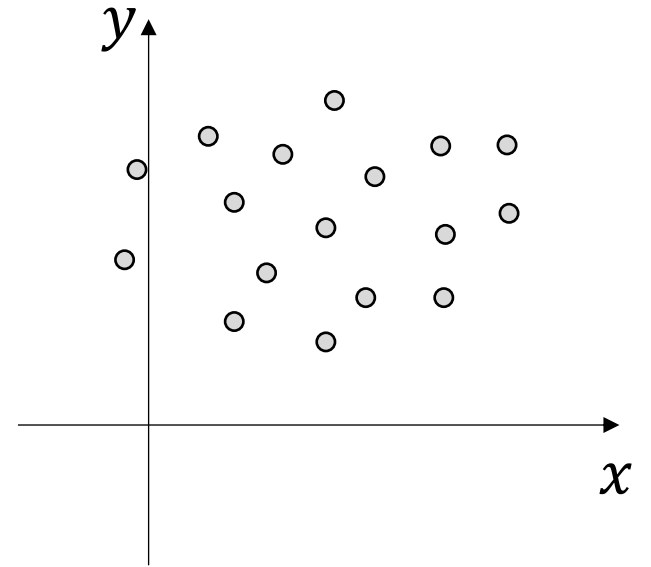
상관 관계



양의 상관관계
positive correlation



음의 상관관계
negative correlation



상관관계 없음
no correlation

공분산(covariance)

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

공분산(covariance)

$$Cov(X, Y) = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

x	y
1	3
2	5
6	4

$$\bar{x} = 3 \quad \bar{y} = 4$$

$$\begin{aligned} Cov(X, Y) &= \frac{1}{3 - 1} [(1 - 3)(3 - 4) + (2 - 3)(5 - 4) + (6 - 3)(4 - 4)] \\ &= \frac{1}{3 - 1} [(-2)(-1) + (-1)(1) + (3)(0)] \\ &= \frac{1}{2} [2 - 1 + 0] = \frac{1}{2} \end{aligned}$$

공분산(covariance)

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

x	y
1	3
2	5
6	4

$$\bar{x} = 3 \quad \bar{y} = 4$$

x	$\bar{x}$	$x - \bar{x}$
1	3	-2
2	3	-1
6	3	3

y	$\bar{y}$	$y - \bar{y}$
3	4	-1
5	4	1
4	4	0

공분산(covariance)

$$\begin{aligned} Cov(\mathbf{X}) &= \frac{1}{n-1} (\mathbf{X} - \bar{\mathbf{X}})^T (\mathbf{X} - \bar{\mathbf{X}}) \\ &= \frac{1}{n-1} \mathbf{X}^T \mathbf{X} \end{aligned}$$

공분산(covariance)

$x - \bar{x}$
-2
-1
3

$y - \bar{y}$
-1
1
0

$$\frac{1}{2} \left[\begin{array}{ccc} -2 & -1 & 3 \\ -1 & 1 & 0 \end{array} \quad \begin{array}{cc} -2 & -1 \\ -1 & 1 \\ 3 & 0 \end{array} \right]$$

$(X - \bar{X})^T$ $(X - \bar{X})$

공분산(covariance)

In [36]:

1 df

Out[36]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222
...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273

506 rows × 14 columns

In [35]:

1 X = df[['CRIM', 'NOX', 'RM', 'AGE', 'TAX']]

In [34]:

1 X

Out[34]:

	CRIM	NOX	RM	AGE	TAX
0	0.00632	0.538	6.575	65.2	296
1	0.02731	0.469	6.421	78.9	242
2	0.02729	0.469	7.185	61.1	242
3	0.03237	0.458	6.998	45.8	222
4	0.06905	0.458	7.147	54.2	222
...	...	...	...	...	...
501	0.06263	0.573	6.593	69.1	273
502	0.04527	0.573	6.120	76.7	273
503	0.06076	0.573	6.976	91.0	273
504	0.10959	0.573	6.794	89.3	273
505	0.04741	0.573	6.030	80.8	273

506 rows × 5 columns

공분산(covariance)

```
1 cov_x = np.cov(X, rowvar=0)
2 print(cov_x)
```

```
[[ 7.39865782e+01  4.19593894e-01 -1.32503785e+00  8.54053223e+01
   8.44821538e+02]
 [ 4.19593894e-01  1.34276357e-02 -2.46034495e-02  2.38592720e+00
   1.30462855e+01]
 [-1.32503785e+00 -2.46034495e-02  4.93670850e-01 -4.75192919e+00
  -3.45834478e+01]
 [ 8.54053223e+01  2.38592720e+00 -4.75192919e+00  7.92358399e+02
   2.40269012e+03]
 [ 8.44821538e+02  1.30462855e+01 -3.45834478e+01  2.40269012e+03
  2.84047595e+04]]
```

```
In [81]: ► 1 cov_x = X.cov()
          2 cov_x
```

Out[81]:

	CRIM	NOX	RM	AGE	TAX
CRIM	73.986578	0.419594	-1.325038	85.405322	844.821538
NOX	0.419594	0.013428	-0.024603	2.385927	13.046286
RM	-1.325038	-0.024603	0.493671	-4.751929	-34.583448
AGE	85.405322	2.385927	-4.751929	792.358399	2402.690122
TAX	844.821538	13.046286	-34.583448	2402.690122	28404.759488

```
1 cov_x = np.cov(X.T)
2 print(cov_x)
```

```
[[ 7.39865782e+01  4.19593894e-01 -1.32503785e+00  8.54053223e+01
   8.44821538e+02]
 [ 4.19593894e-01  1.34276357e-02 -2.46034495e-02  2.38592720e+00
   1.30462855e+01]
 [-1.32503785e+00 -2.46034495e-02  4.93670850e-01 -4.75192919e+00
  -3.45834478e+01]
 [ 8.54053223e+01  2.38592720e+00 -4.75192919e+00  7.92358399e+02
   2.40269012e+03]
 [ 8.44821538e+02  1.30462855e+01 -3.45834478e+01  2.40269012e+03
  2.84047595e+04]]
```

공분산(covariance)

$$-\infty < Cov(X, Y) < \infty$$

In [81]: ▶

```
1 cov_x = X.cov()  
2 cov_x
```

Out[81]:

	CRIM	NOX	RM	AGE	TAX
CRIM	73.986578	0.419594	-1.325038	85.405322	844.821538
NOX	0.419594	0.013428	-0.024603	2.385927	13.046286
RM	-1.325038	-0.024603	0.493671	-4.751929	-34.583448
AGE	85.405322	2.385927	-4.751929	792.358399	2402.690122
TAX	844.821538	13.046286	-34.583448	2402.690122	28404.759488

공분산은 데이터 값의 단위에 영향을 받는다

피어슨 상관계수(Pearson Correlation Coefficient)

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$


X의 표준편차

피어슨 상관계수(Pearson Correlation Coefficient)

$$-1 \leq \text{Corr}(X, Y) \leq 1$$

$-1 \leq \text{Corr}(X, Y) < -0.6$ 강한 음의 상관 관계

$-0.6 < \text{Corr}(X, Y) < 0$ 약한 음의 상관 관계

$0 < \text{Corr}(X, Y) < 0.6$ 약한 양의 상관 관계

$0.6 < \text{Corr}(X, Y) \leq 1$ 강한 양의 상관 관계

피어슨 상관계수(Pearson Correlation Coefficient)

X, Y 가 서로 상관관계가 없으면 $\Rightarrow \text{Corr}(X, Y) = 0$

$\text{Corr}(X, Y) = 0 \not\Rightarrow X, Y$ 가 서로 상관관계가 없다

피어슨 상관계수(Pearson Correlation Coefficient)

In [82]: ▶

```
1 corr_x = np.corrcoef(X.T)
2 print(corr_x)
```

```
[[ 1.          0.42097171 -0.2192467  0.35273425  0.58276431]
 [ 0.42097171  1.          -0.30218819  0.7314701  0.6680232 ]
 [-0.2192467  -0.30218819  1.          -0.24026493 -0.29204783]
 [ 0.35273425  0.7314701  -0.24026493  1.          0.50645559]
 [ 0.58276431  0.6680232  -0.29204783  0.50645559  1.          ]]
```

In [84]: ▶

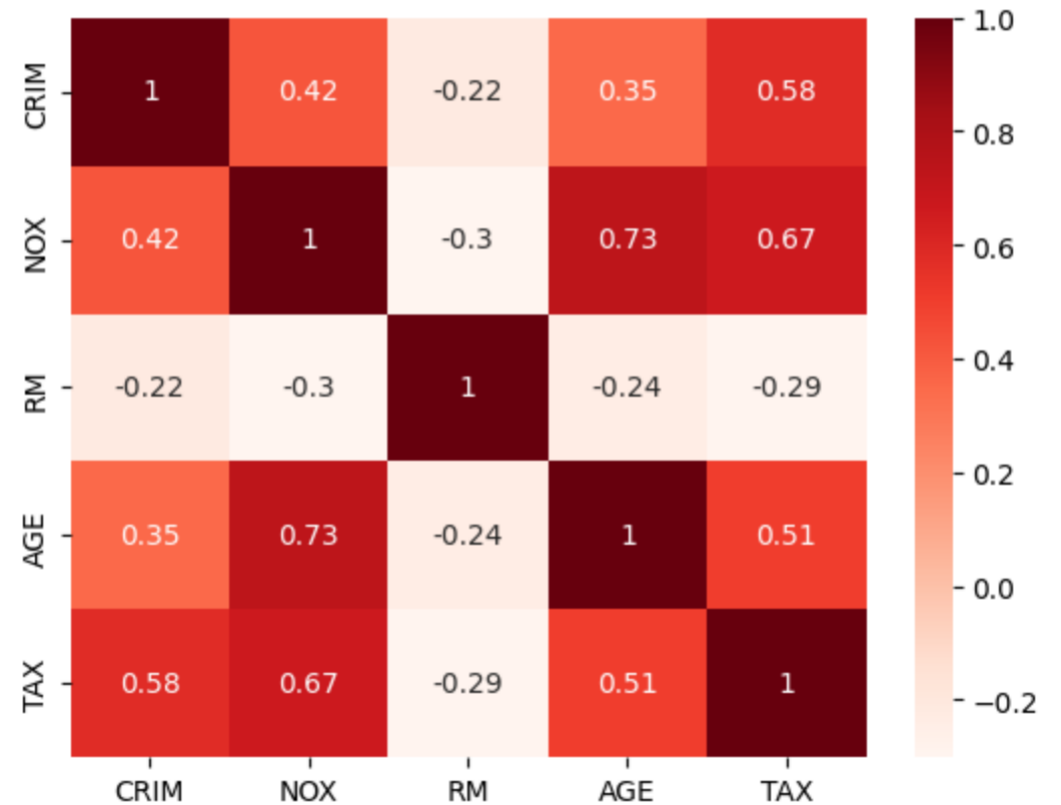
```
1 corr_x = X.corr()
2 corr_x
```

Out[84]:

	CRIM	NOX	RM	AGE	TAX
CRIM	1.000000	0.420972	-0.219247	0.352734	0.582764
NOX	0.420972	1.000000	-0.302188	0.731470	0.668023
RM	-0.219247	-0.302188	1.000000	-0.240265	-0.292048
AGE	0.352734	0.731470	-0.240265	1.000000	0.506456
TAX	0.582764	0.668023	-0.292048	0.506456	1.000000

```
1 import seaborn as sns
2
3 sns.heatmap(corr_x, annot=True, cmap='Reds')
```

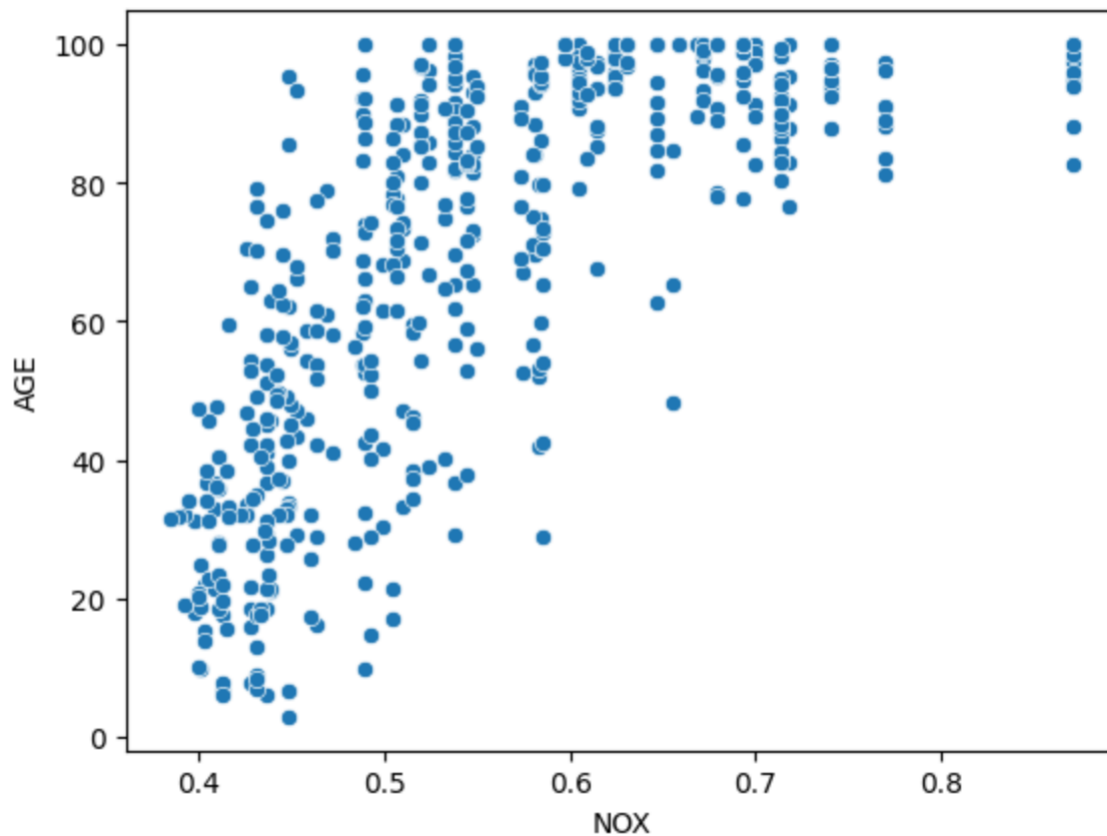
<Axes: >



피어슨 상관계수(Pearson Correlation Coefficient)

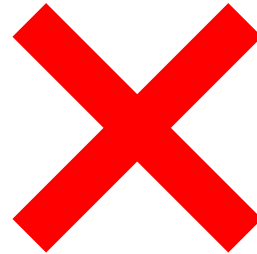
```
In [89]: 1 sns.scatterplot(data=df, x="NOX", y='AGE')
```

```
Out[89]: <Axes: xlabel='NOX', ylabel='AGE'>
```



피어슨 상관계수(Pearson Correlation Coefficient)

상관 관계가 존재하면 인과 관계가 존재한다?



감사합니다.

Q & A